Raise Borer

31/10/2019

Raise Borer Applicability I

*Raise excavated by raise boring is circular & excavation is free from drilling & blasting.

*Both, upper level and lower level should be available for excavating raise by raise borer.

*Raise boring can be applied to soft as well as hard ground.

*Bored raise can be drilled vertically or inclined.

Raise Borer Applicability II

*By raise boring, raise up to 910m long & 30 degree incline can be excavated.

*Incline raise up to 45 degree can be excavated by raise boring confortably.

*Raise from 0.9m to 3.7m diameter can be excavated by raise boring with out reaming.

*Raise boring can be effectively applied in relatively poor ground.

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Drill Direction I

*Decision on drilling direction depends on availability of access to raise site, ease in transportation & installation of raise borer, speed, energy required & over all economy of operation.

*Space & facility available at each end of raise mainly govern choice of drilling method.

*If only one end of raise has adequate rooms, machine must be put there.

Drill Direction II

*Location of borer is placed such a way to create least disturbance in mine.

*Generally, drilling down pilot hole & reaming up is followed, because this is cheapest & reliable option.

*Rock quality & its predicted response to drilling can influence choice of direction.

Drill Direction III

*Raise boring machines operate on principle of first drilling a small pilot hole & then reaming hole in one or more stages, to achieve desired size.

*Raise boring machine is set up at upper level.

*A pilot hole of 225 to 250mm dia is first drilled down to lower level.

*Pilot hole also provide information about type of strata to be encountered & helps drilling raise accurately.

Drill Direction IV

*In case of large deviation, pilot hole has to be abandoned & fresh hole to be drilled.

*A large reamer bit is put on at bottom of drill rod and raise is reamed at desired diameter.

*Reverse process can also be adopted i.e. first driving pilot upward from lower level towards upper level and then reaming it from upper level to bottom level.

*Some time single stage drilling of raise is also adopted.

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Raise Boring Machine I

*Raise boring are originally designed for shoft rock.

*Now machines are available for hard rock.

*Different raise boring machines are available for different rock conditions, nominal sizes & length of finished hole.

Raise Boring Machine II

Machine specification includes following

*Pilot hole diameter.

*Drill pipe diameter

*Reaming torque

*Type of drive (AC, DC, hydraulic)

*RPM

Raise Boring Machine III

*Power kw

*Stages & type gear reduction & RPM

*Pilot hole thrust

*Feed rate

*Reaming pull

Raise Boring Machine IV

*Size of drill pipe

*Base plate.

*Derrick dimension with characteristic

*Type of transport / erector

*Water consumption

*Air consumption

Raise Boring Machine V

*Rig is itself a rigid plate & structure, on which drilling, hydraulic equipment are assembled.

*Drill rig subjected to wear of bearing & drilling tools, vibration on uneven surface, rubbing of drill pipes against drill hole

*Raise boring machines have provision of drilling in either way.

Raise Boring Machine VI

*Raise boring machinea are designed for drilling in soft & hard ground, both.

*Raise boring machine can be dis assembled in to smaller component, transported by underground service type vehicle. to new working place & assembled again.

*Raise boring machine also comes with crawler, wheel or skid mount, rail wheel for haulage.

Raise Boring Machine VII

*One operator with one helper is sufficient to operate raise boring machine.

*Operation & maintenance crew are to be specifically trained for operating raise boring machine.

*Place, from where drilling of pilot hole has to start, should be cleaned, made free from loose, checked for misfire & leveled with help of PCC.

*This arrangement shall help to set drill machine for accurate drilling.

Drill String I

*Pilot hole is drilled through stem with stabilizer & conventional drill bit.

*Drill rod has outer / inner diameter in range of 20.3 to 30.4mm (8 to 12.7/8 inch) to 12.06 mm to 13.1mm (4.3/4 to 5.7/16 inch).

*Length of section for underground application is usually 1.5m, dictated by clearance restriction.

*Connection are made with tapered thread joints that have to be kept absolutely cleaned.

Drill String II

*At time of drilling pilot hole, drill strings are added one by one & at time of reaming, drill strings are removed.

*Tension on stem has to be adjusted to allow starting hole with out hitting variable dynamic load acting on drilling component.

*String (stem) is longest at this stage of drilling & there fore, prone to maximum elongation & twist.

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Drill Stabilizer I

*Spacing & number of stabilizer varies based on rock condition, drill string, drilling pressure.

*Purpose of stabilizer is to assure directional accuracy of pilot hole.

*They are larger diameter than drill roads, will strap of tungsten carbide welded vertically or spirally on out side, brining outer dimension close to hole diameter.

*These stabilizers are used for reaming up.

*Stabilizer for pilot holes & reaming down, have more clearance, allowing cutting to pass.

Drill Stabilizer II

*Stabilizer follows pilot bit & mounted on drill string.

*Some time number of stabilizers are used distributed on drill string.

*Once pilot hole penetrates to mine opening, pilot bit & stabilizers removed.

*After necessary servicing, stabilizer is mounted on drill stem & reamer is placed back.

Drill Speed & Bit I

*Standard analysis is required & is usually performed by computer, taking in to account necessary balancing tools, achieved by optimization of cutter.

*Rotational speed for pilot hole drilling varies from 35-72 rpm and for reaming, speed is 10-20 rpm.

*Presser on 250mm (9.8 inch) pilot bit is 13.5t (30,000 lb) and for 380mm (15 inch) it is 57t (125000 lb).

*Pressure on reaming bit varies from 9t (20000 lb) for 1.2m (48 inch bit) to 15t (36000 lb) for 1.5m (60 inch) bit.

Drill Speed & Bit II

*Starting hole (collaring) is a critical operation.

*Damage to cutter / bit body can occur due to uneven surface.

*Design of reaming depends on final size of hole & number of reaming stage.

*Different raise reamers are used for different diameter holes & raise.

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Drill Speed & Bit III

*Some time, catenation are attached to primary body of cutter.

*By this, an increase of final diameter is achieved by addicting more cutter in same plane as cutters on primary body itself.

Drilling Performance I

*Drilling performance is measured by rate of penetration.

*Drilling performance for a particular rock greatly depends on bit design.

*Cutting action has to match rock penetration to optimize drilling rate.

*Design of reaming head influence raise boring process while reaming.

*An analysis of presently used cutter can help in design improvement of cutting standard.

Drilling Performance II

*Independent variable characteristic of drilling performance depends on tensile & shear strength of rock.

*Drilling tool should break rock primarily in tension & shear, because strength or rock in these made of loading are minimal.

*Rock ductility & abrasibility also influence drillability of rock

*Strong but brittle rock has better drilling response than weak, clay rock.

Drilling Performance III

*Abrasibility depends on mineral content of rock & is responsible for were of drilling tool element in contact with rock face & cutting.

*Properties of rock mass, joints, faults, structured features also influence drill response.

Drilling Performance IV

*In raise boring, this situation is encountered during drilling of pilot hole.

*Drilling technology with compressed air, air & foam, drilling fluids is used for cutting removals.

*During reaming operation, cutting falls down the hole.

*Drilling load should be at level required by cutter to penetrate & create fracture.

*Once thresh hold strength rock is reached, further increase in drilling load does greater depth penetration to local , but not in direct proportion.

Drilling Performance IV

$D=Wd$

D> depth of penetration by drilling bit

W> approved load

d> load exponent (1.5 in soft rock, 1 in hard rock)

*Operator measures rate of penetration over several minutes.

*Load on drill bit is adjusted through several trials to find optimum load, above which load, rate of penetration does not noticeably increases.

*Rotary speed also effects penetration rate.

*Higher rotary speed, faster rate of penetration.

Drilling Cuttings I

*Drill cutting gets deposited around hole collar & they should be removed immediately from hole collar.

*Drill cutting should not get back in to hole in any condition.

*Drill hole casing is kept little above floor level to prevent reverse flow.

*Dry drill cutting are to be removed by hand shovel.

*When raise boring machine is located at lower level, cutting from drilling operation drops down by gravity to lower level.

Drilling Cuttings II

*Drill cutting are to be collected in hopper then to chute / pipe line for final disposal.

*Layer of cutting on uncut rock reduces penetration up to 40%.

*Drill cutting should be flushed away as early as possible.

*Energy of drilling is spent on regrinding cutting rather than penetrating intact rock.

Raise Support I

*In strong rock without fracture usually no support is required.

*If support is required, shotcrete, rock bolts, with without wire mesh or steel lining can be applied.

*Shotcrete can be placed remotely or by worker from cage.

*Rock bolts are to be installed manually.

*Steel lining, required in unstable formation, is installed remotely, directly behind reamer.

Raise Borer I

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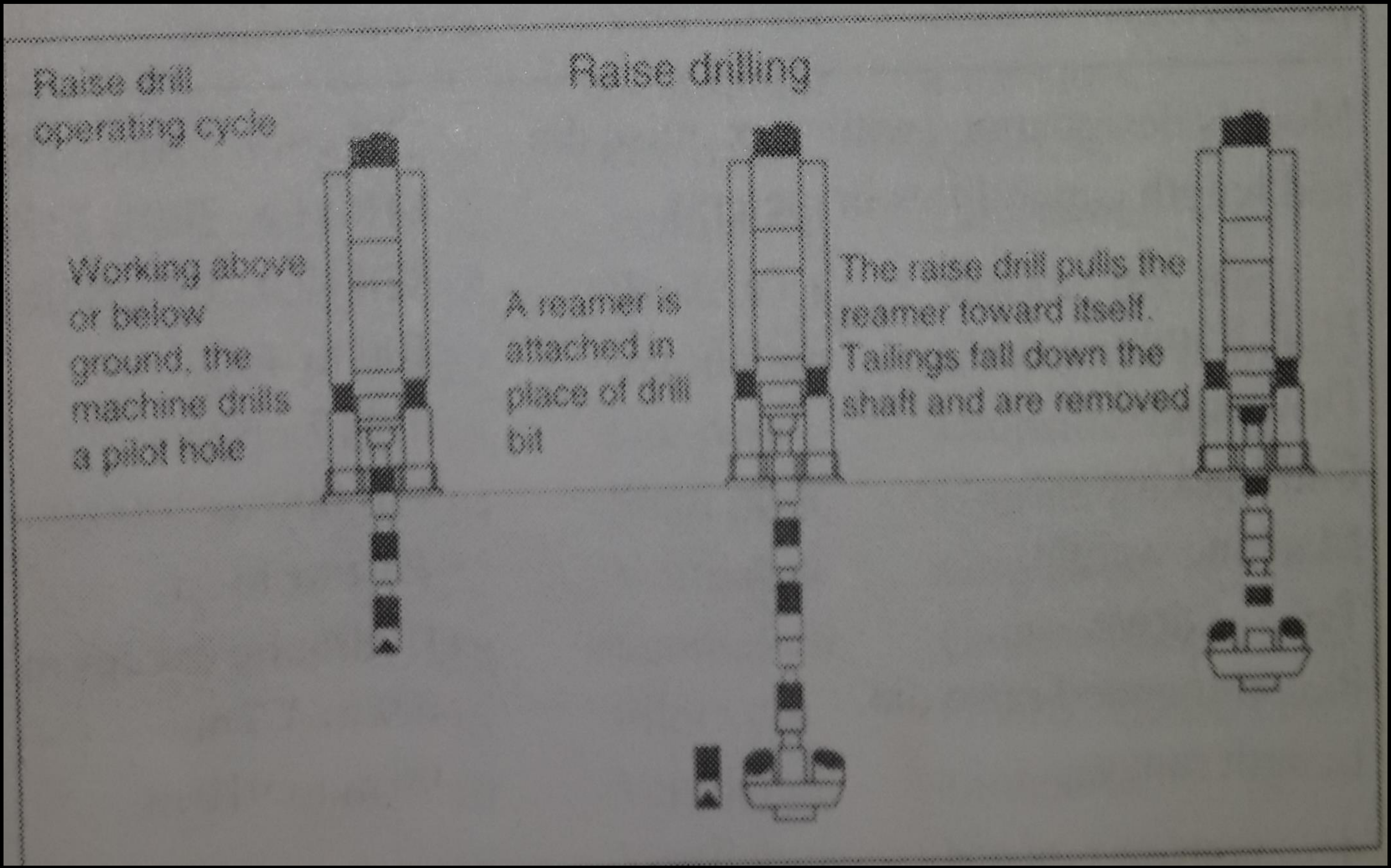
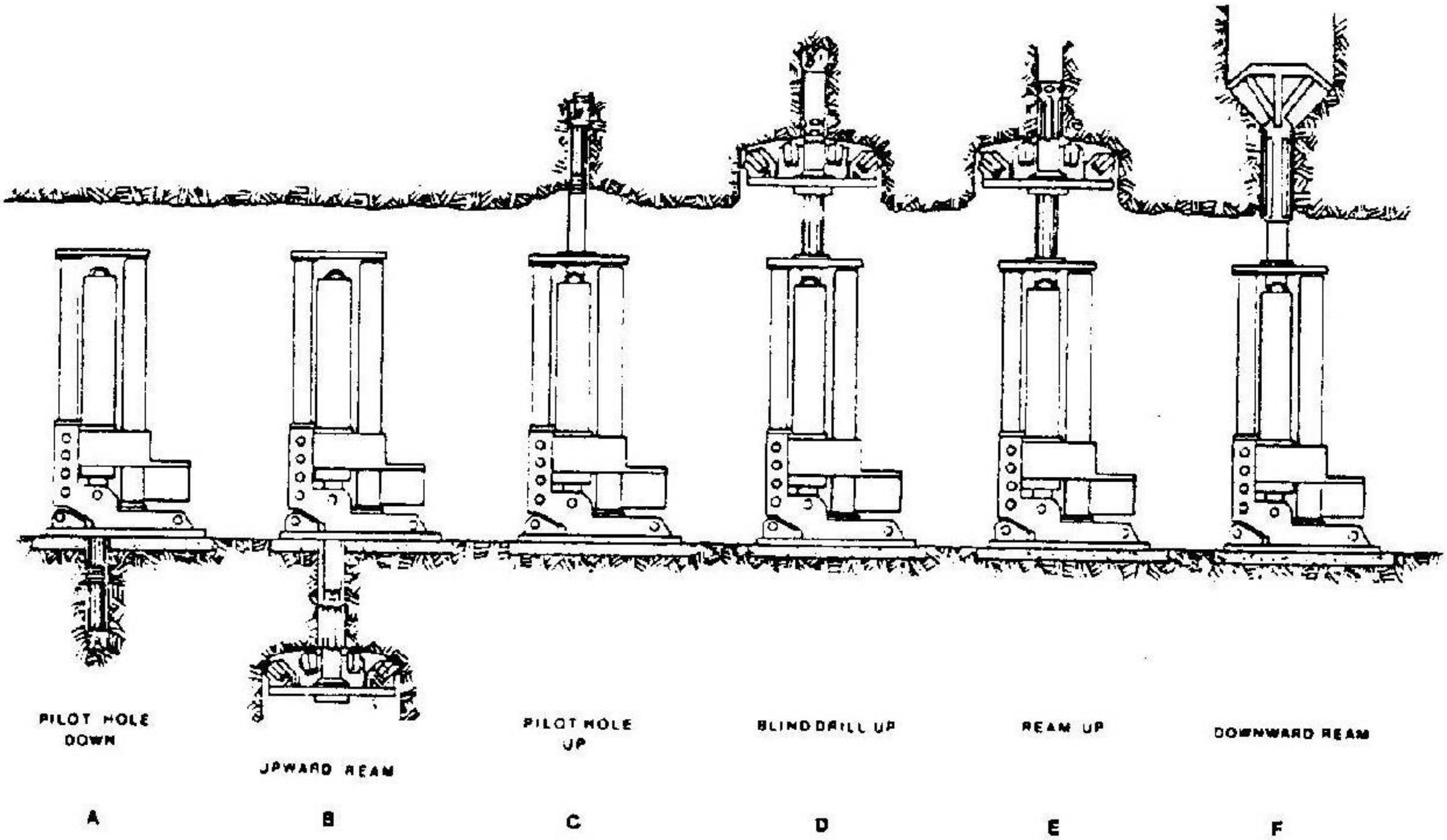


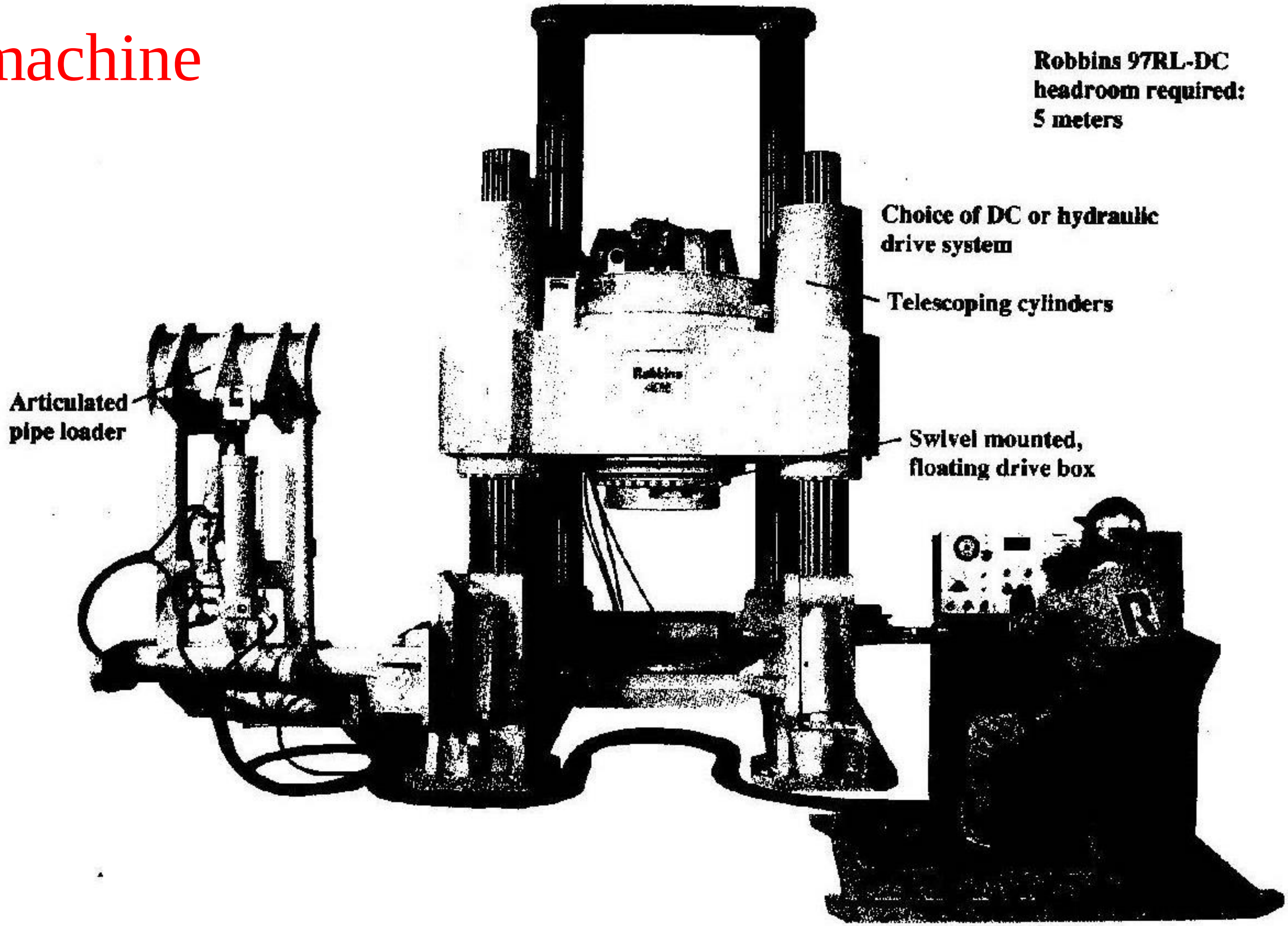
Raise Boring

Modes of operation in raise boring



Raise Boring

Robbins drill machine



Raise Borer I

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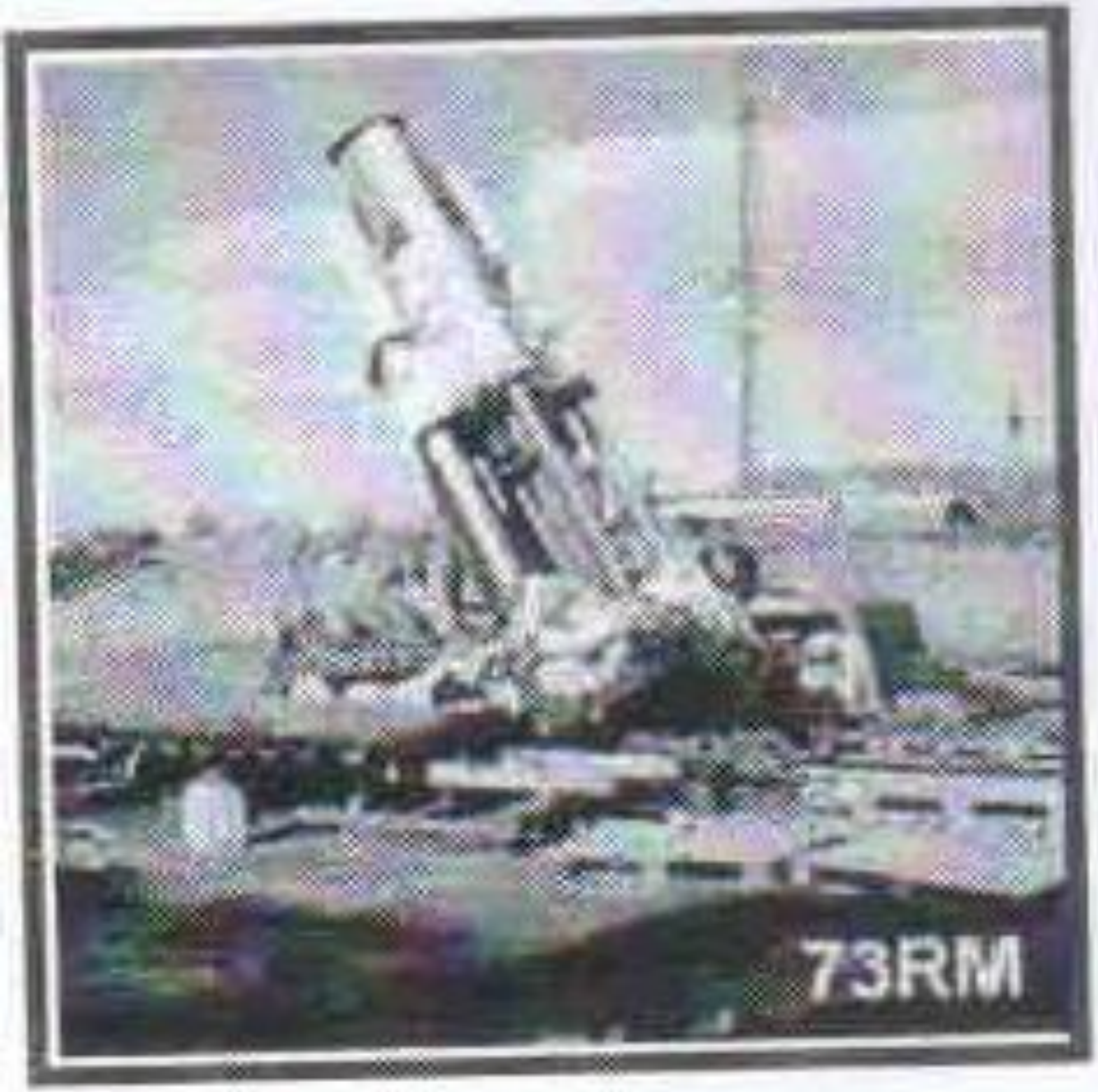
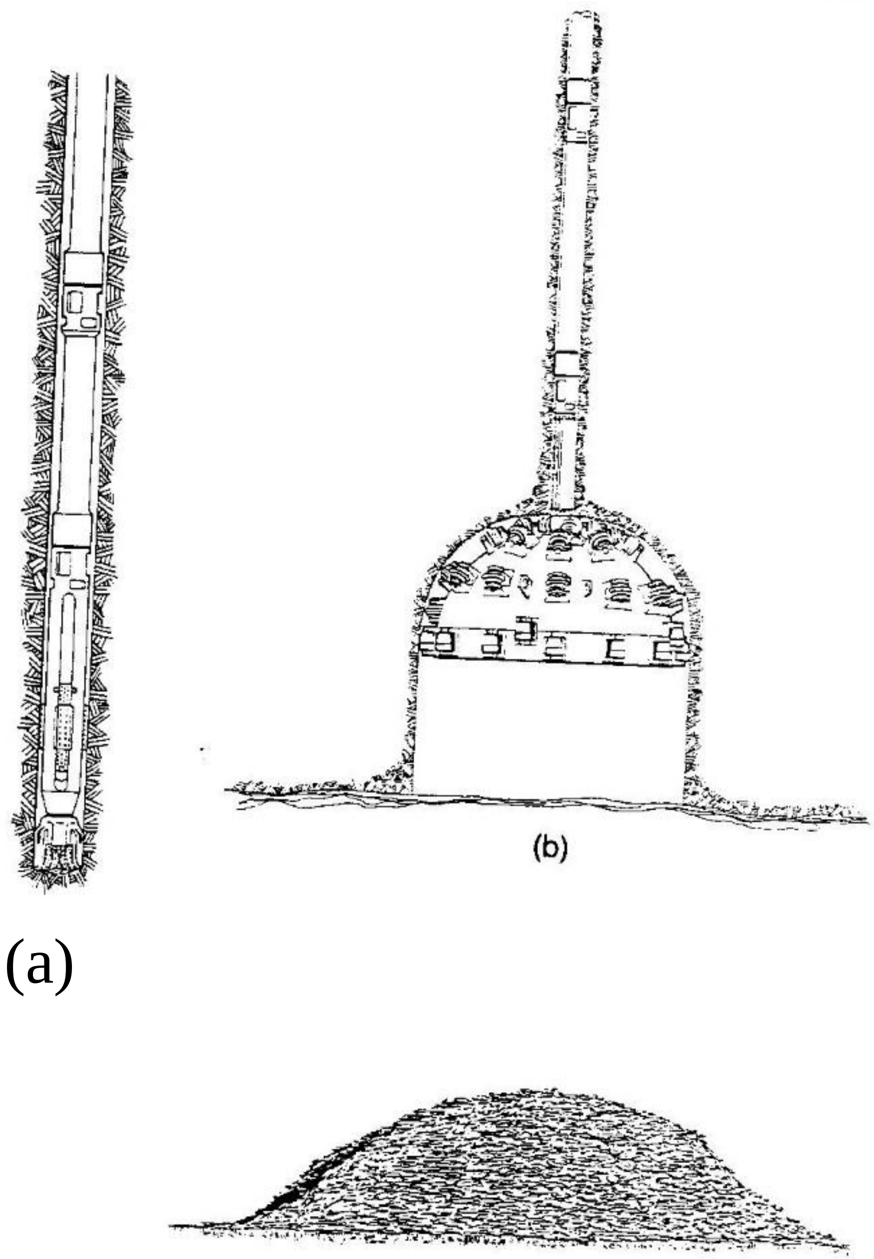


Raise Boring

CONSTRUCTION OF



Pilot hole (a) drilling & (b) reaming

Raise Boring

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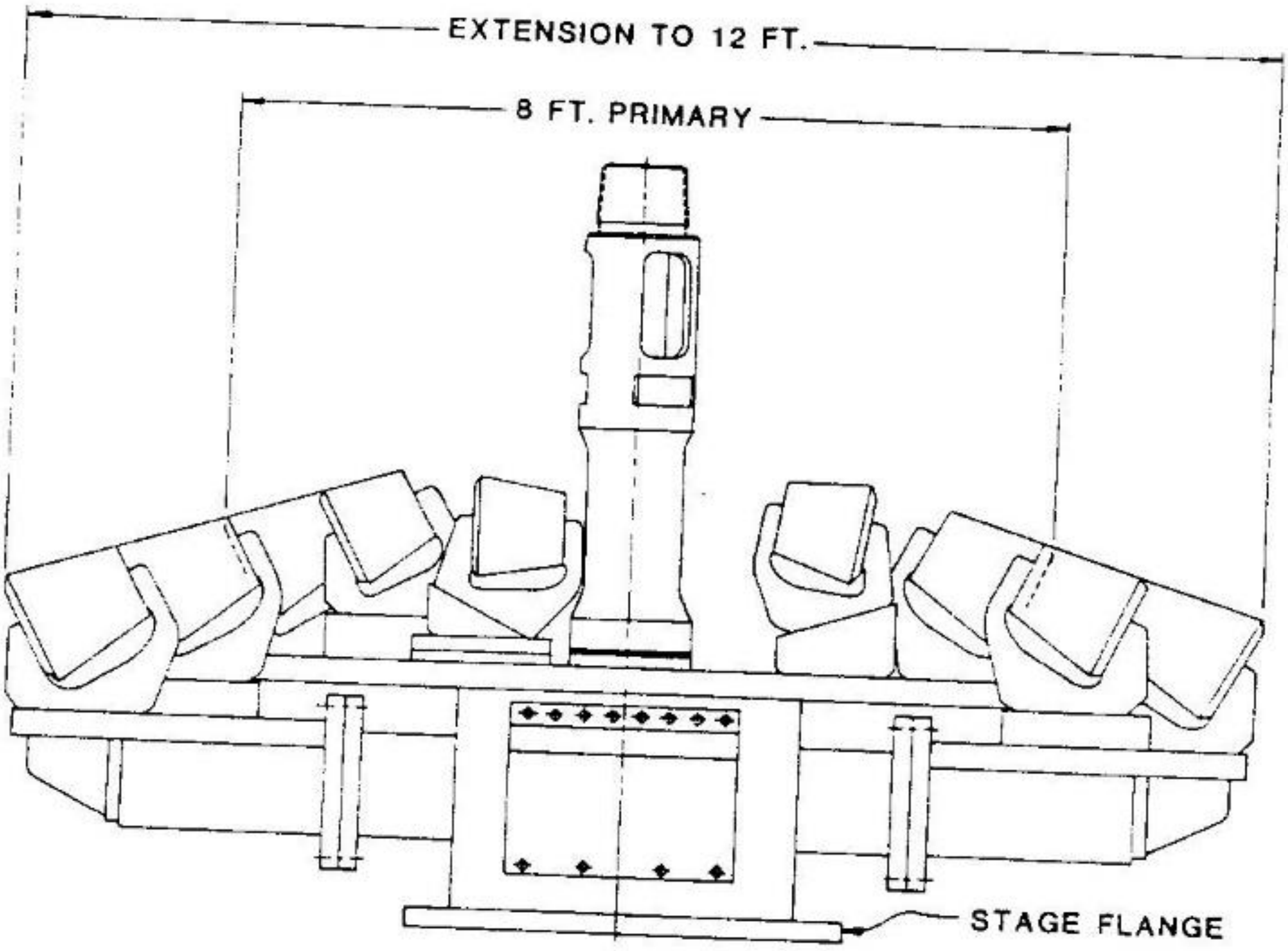

Fig. 17.4.64. Raise bit with extensions. Conversion factor: 1 ft = 0.3048 m.

* Raise Boring

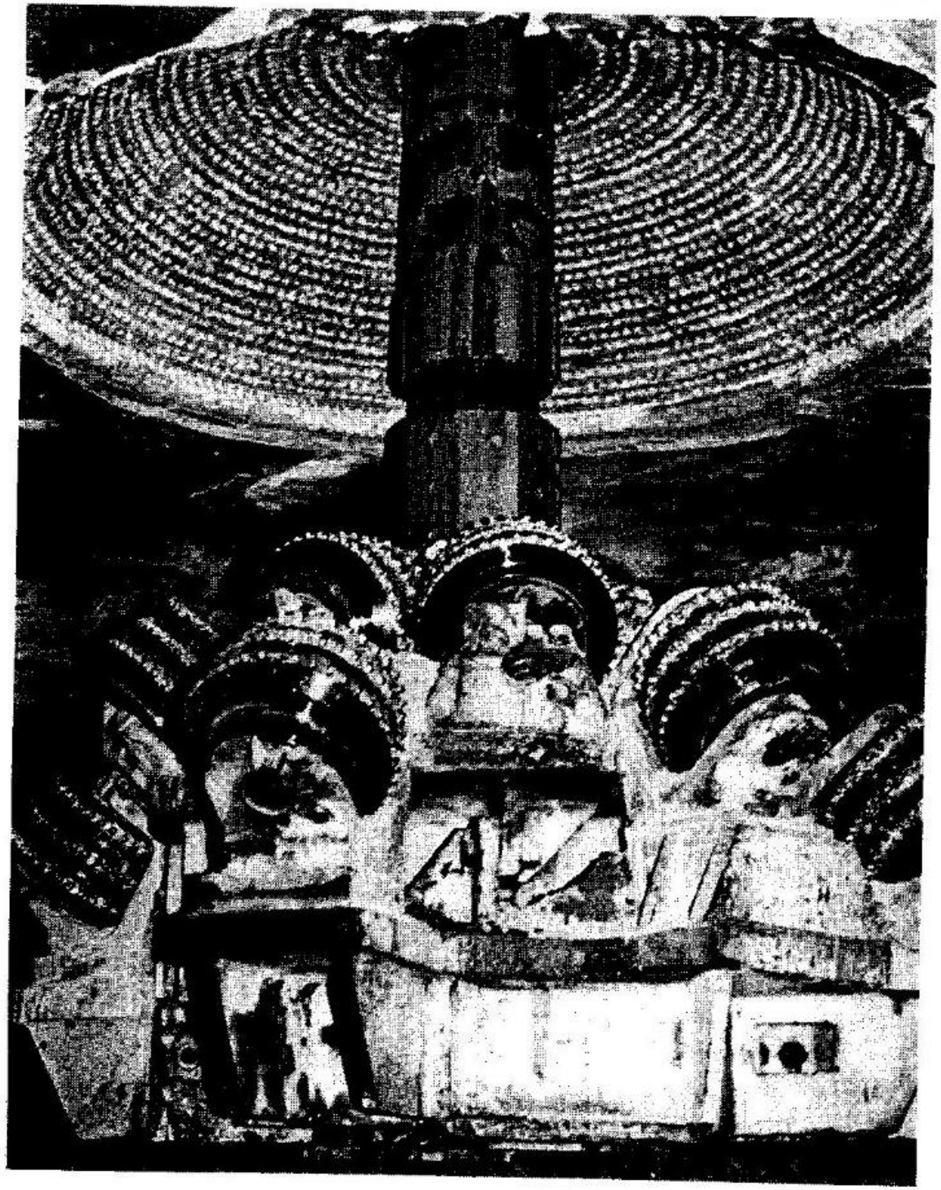

Fig. 17.4.65. Top section of 16-ft (4.8-m) diameter reamer breaking through the surface after 1490 ft (453 m) of reaming.

* Stabilizer

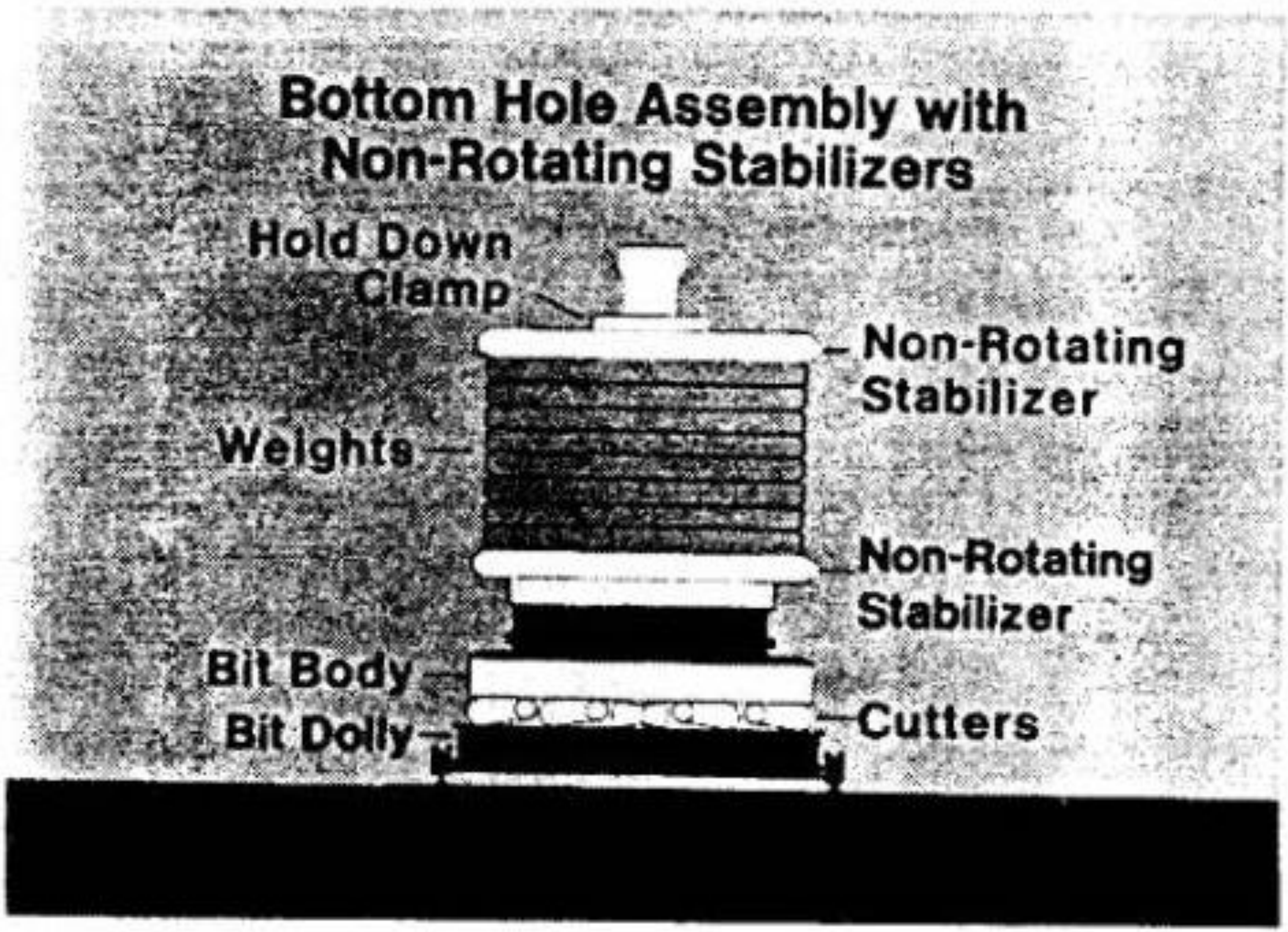

Fig. 17.4.71. 14-ft (4.3-m) diameter bottom-hole assembly.

Borpak I

*This latest development in raise boring.

*This unit is self driven & mounted on crawler track.

*This machine is used for blind raising.

*Drilling is done from bottom to top (from lower level to upper level).

*Boring up ward starts from lower level through launching tube.

*After header has penetrated a few meters in rock, gripper hold body of borer, while head rotates & bores rock.

Borpak II

*Raise diameter is 1.2m to 3m.

*Maximum length of raise is 300m.

*operation of Borpak is similar to tunnel boring machine.

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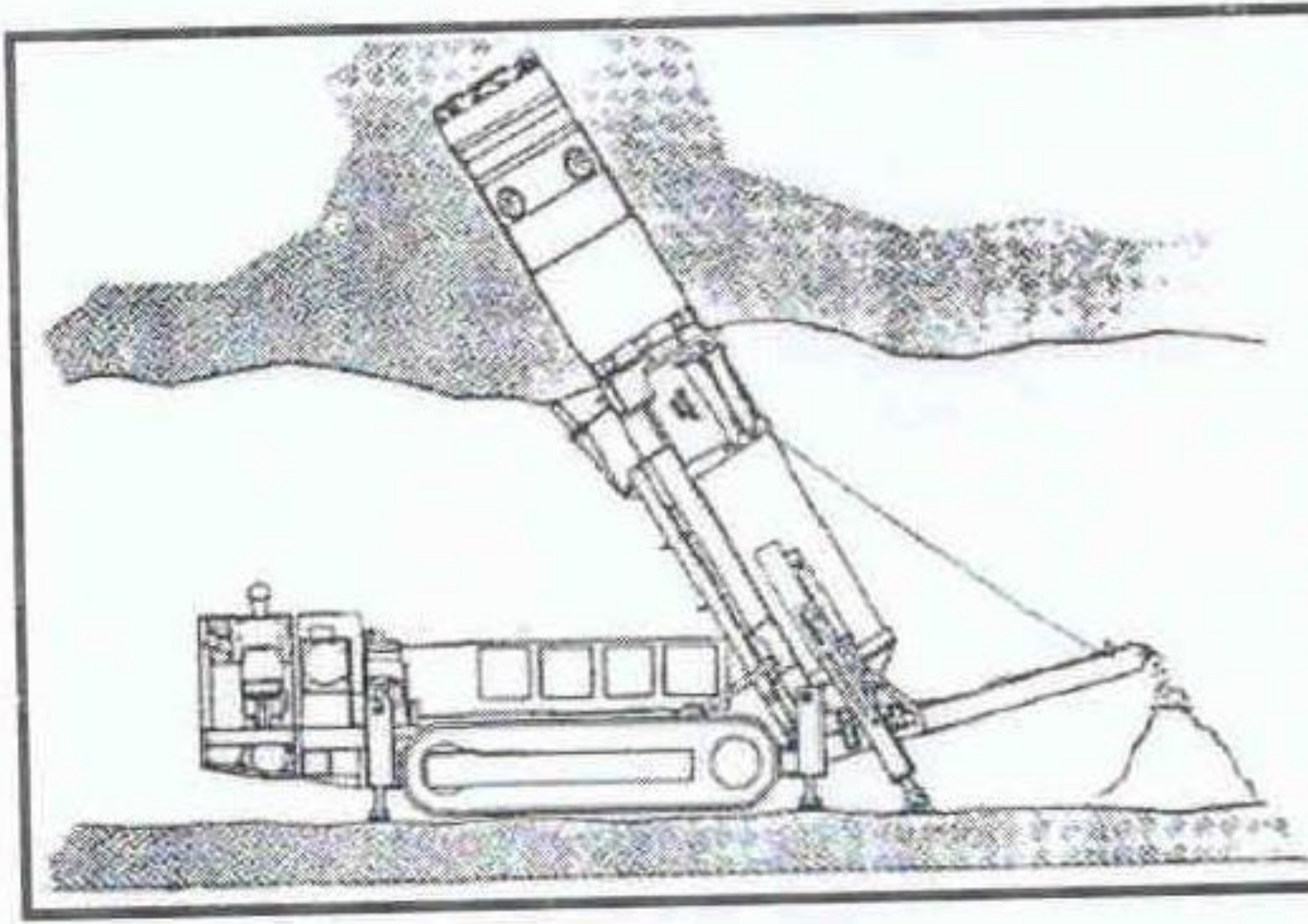
Borpak I



.11 BorPak – raise boring machine.

Raise Borer I

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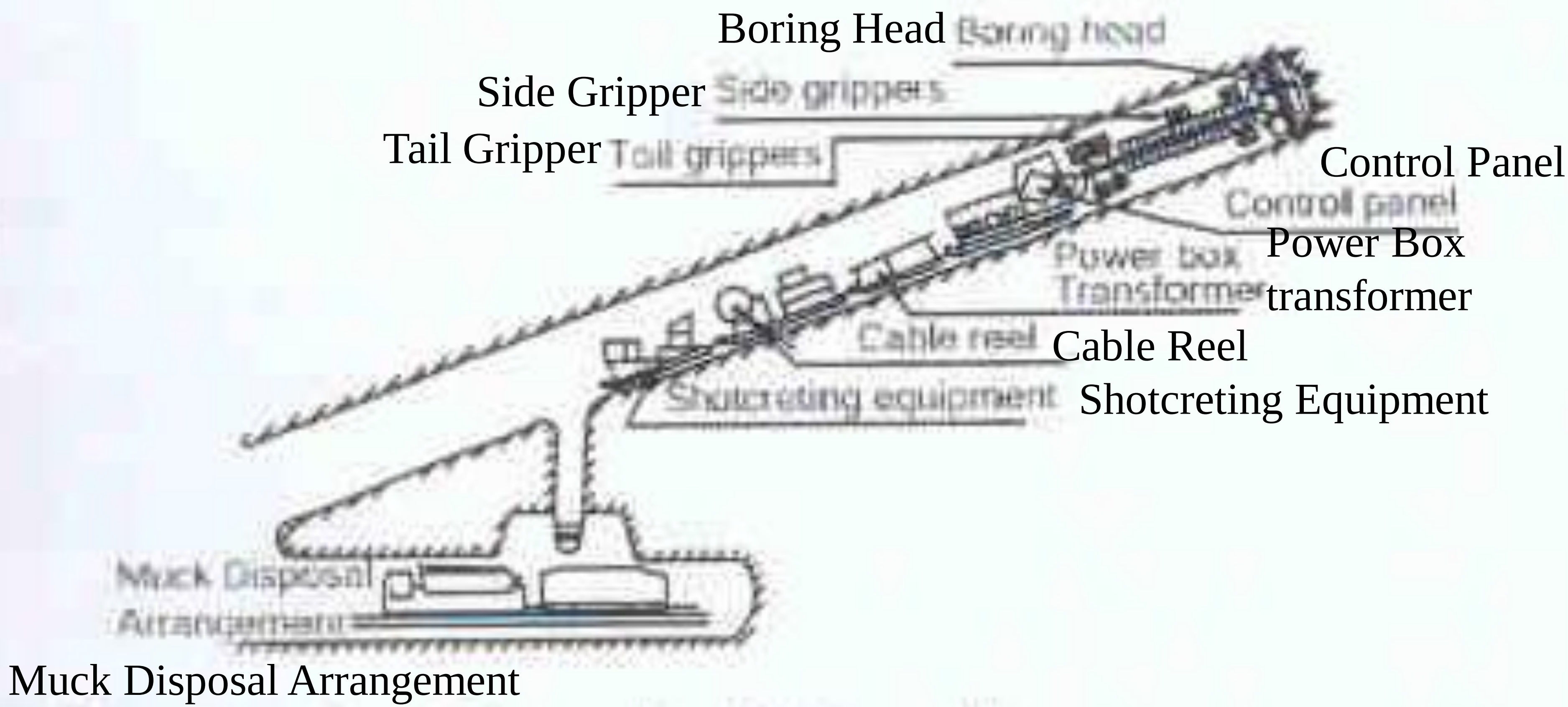

(f) Application of borer for driving inclines

Advantages I

*No worker need to be present at raise face.

*There is no human exposure in to raise at time of raising.

*This reduces exposure of men to hazards condition.

*Reducing man power increases cost effectiveness (cost reduction).

*For cost effectiveness, raise boring machine should not be left ideal.

*Continuous boring assures faster completion of raises.

*Boring does not disturb surrounding rocks.

Advantages II

*There is no need of support or substantial reduction in support requirement.

*Perfectly rounded shape makes excavation inherently stable.

*Its smooth wall reduces resistance for ventilation as well as for passing ore.

*Process of raising is non cyclic, hence faster.

*Process of raising is economic, provided raise boring machines remains engaged with out time loss.

Disadvantages I

*General requirement of rock condition be similar with in length of raise bore.

*Drilling in hard rock is slow & cost rapidly increases.

*Raise bores rock has to be very weak & reasonably stable.

*Initial cost of rig is very high.

Reference I

*SME mining Engineering Hand Book, Hartman, 2nd Edition, Volume 2, Page-1624 to 1627.

*Surface and Underground excavations, Techniques and Equipment. Ratan Raj Tatiya, Page 347 to 350.